Reference Event 1: The Ransomware Attack on JBS, May and June 2021

Characterization Against the Food and Agriculture Sector Design Basis

DBA-FA-RE-1 — Reference Event Instrument, DBA-FA Series

Lineage: Issued under the Food and Agriculture Sector Framework v0.5; DBA-MA Parent Framework; conferred on the proposal of Food Processing and Manufacturing (DBA-FA-FP-FC2)

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the ransomware attack on JBS of May and June 2021 back through DBA-FA-FP-FC2 (*Food Processing and Manufacturing*), whose category of a cyber attack on the plant's systems it characterizes (§5.5), at the class's first reference facility, the large beef or pork slaughter and fabrication plant (§1.3). It asks whether the class's design basis anticipated what happened, and returns what it finds.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Food and Agriculture Sector Framework v0.5, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 14 October 2026, on the proposal of *Food Processing and Manufacturing*. It is the series' first reference event instrument. Section references of the bare form §N are to that class at v0.1, the version this instrument read.

The scope runs from the company's discovery of the attack on 30 May 2021, through the stop of its plants and their return, to its statement of 9 June that it had paid a ransom. The record is closed.

No individual is named. The company, its subsidiary Pilgrim's and the public bodies are named as institutions. The Federal Bureau of Investigation attributed the attack to the criminal ransomware operation known as REvil and Sodinokibi; the attribution is the Bureau's, and this instrument makes no finding of attribution and none of fault.

The plants and the company systems they depend on lie within this class; the cattle and hogs held upstream are Agricultural Production's. The Foundational Definitions are cited by designation, FD-REG holding their versions: FD-BL-D2 (*Indication Integrity*) bears on Section 6.4.

2. Sources and Attribution

2.1 Sources and Their Limit

The primary sources are the company's statements of 31 May, 3 June and 9 June 2021 and the Bureau's statement of 2 June 2021. The effect on national slaughter and the company's share of beef capacity are from Bloomberg's report of 2 June 2021, which gives the share from an investor report of another processor, and from the Department of Agriculture's slaughter estimates as reported. No non-public material is used.

2.2 Where the Sources Differ

They do not differ. The company's share of national beef capacity, about 23 percent, rests on a competitor's investor materials as reported, not on a public body's figure, and is stated as such.

3. The Event

3.1 The Company and Its Plants

JBS is among the largest meat processors in the world. Its United States beef plants, with its pork plants and the poultry plants of its subsidiary Pilgrim's, ran on business systems served from a common set of servers for North America and Australia.

3.2 The Attack and the Company's Response

On Sunday 30 May 2021 the company found that it was the target of an organized cybersecurity attack affecting some of the servers supporting its North American and Australian information systems. It "took immediate action, suspending all affected systems", notified the authorities and brought in outside specialists. It stated that its backup servers were not affected (statement of 31 May).

3.3 The Stop

On 1 June all of the company's United States beef plants stopped or cancelled shifts, among them plants in Utah, Texas, Wisconsin, Nebraska, Iowa and Colorado; its pork plants cut back; two poultry plants closed; a beef plant in Alberta was idled; and slaughter in Australia stopped. National cattle slaughter that day fell 22 percent and hog slaughter 20 percent from a week earlier, on the Department of Agriculture's estimates as reported (Bloomberg). The Department encouraged other processors to add capacity where they could, as reported.

3.4 The Return

On 2 June the company said the vast majority of its facilities had resumed operations. On 3 June it stated that all its facilities worldwide were fully operational, that it had limited the loss of production to less than one day's worth, that its encrypted backup servers, "which were not infected during the attack", had allowed a faster return, and that "the criminals were never able to access our core systems". On 9 June it stated that it had paid the equivalent of $11 million in ransom, at a time when the vast majority of its facilities were operational, "to mitigate any unforeseen issues related to the attack and ensure no data was exfiltrated".

3.5 The Attribution

On 2 June the Bureau stated: "We have attributed the JBS attack to REvil and Sodinokibi and are working diligently to bring the threat actors to justice."

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-3) and effect elements (E-1 to E-4), as Table 4-1 sets out. Composed, they are the class's cyber attack on the plant's systems (§5.5) at the large beef plant (§1.3), in Chain U across every plant of one company at once, without reaching the systems that control the process.

Table 4-1. Event elements and their mapping. Section references are to *Food Processing and Manufacturing* v0.1.

No.	What happened	Class provision	Section
I-1	Attack on shared servers for North America and Australia	Cyber attack on the plant's systems (§5.5)	3.2
I-2	Company suspends all affected systems	Orderly stop as a protective action (§6.7)	3.2
I-3	Backups not reached; core systems not reached	Manual mode; trusted records (§5.5)	3.4
E-1	Every United States beef plant of the company stops within a day	Supply share (§3.3); Chain U (§3.2)	3.3
E-2	National cattle slaughter down 22 percent, hogs 20 percent	Supply share; perishable inflow (§2.1, §3.3)	3.3
E-3	Most plants back in two days; under one day's production lost	Time to restart (§3.3)	3.4
E-4	Other processors asked to add capacity	Substitution of capacity (§4.5)	3.3

5. Tier Determination

5.1 The Four Characteristics

Intentionality is present: the attack was deliberate, a crime for money on the Bureau's account. Persistence is not determined: the sources read do not say how long the attackers were inside before the attack was found. Precision is present in breadth, the servers common to two continents' plants, and absent in any timing to a moment of scarcity. Denial is not present: the attack did not reach the core systems or the backups, and the plants returned within days.

5.2 Determination

The event does not meet the threshold for the military-action extension. It was a crime for money by a criminal group, attributed to no state in the sources read, in peacetime, unaccompanied by any other act. It is a criminal ransomware attack realizing the class's §5.5 in Chain U across one company's plants, and the single-company form of the coordinated cyber scenario of §6.5, which it did not reach. That is the tier determination returned for the DBA Reference Event Register. No impacted sector beyond Food and Agriculture is proposed.

6. Characterization Against the Design Basis

6.1 §5.5 The Class's Account of the Event

Held. The account holds in every figure checked against the company's and the Bureau's statements: the discovery on 30 May, the servers for North America and Australia, the stop in three countries, the Bureau's attribution, less than one day's production lost, the core systems not reached, and the ransom of $11 million. The conferral note asked whether a primary source gives the company's share of beef capacity. None was found: the figure of about 23 percent comes from another processor's investor materials as reported by Bloomberg, and the class may state it so. That is a refinement, not an erratum.

6.2 §3.3 The Supply Share

Held, with a gap. The class states the supply share for a plant. The event stopped every beef plant of one company in a day because they shared business systems, and national cattle slaughter fell by about a fifth. The unit of the supply share was the shared system, not the plant. Finding F-1.

6.3 §5.5 and §6.7 The Company's Containment Stop

Held. The company suspended the affected systems at once, and the plants stopped as a consequence. The class's statement of an orderly stop as a protective action, with its cost named, reads it; the cost here fell on the supply share and on the producers upstream.

6.4 §5.5 and FD-BL-D2 Records, Backups and the Manual Mode

Held, with a gap. The class requires that a plant can process, label and trace without its business systems and that it not release product on records it cannot trust. The event's short duration rested on two things the company reported and the plants did not control: the backups were not reached, and the core systems were not reached. The class does not say what may be credited for restoration. Finding F-2.

6.5 §7.2 The Process Controls and Protective Functions

Not exercised. Nothing in the sources read shows the attack reaching the plants' process controls, kill steps or refrigeration protection; the separation the class requires was not tested.

6.6 §4.5 Substitution of Capacity

Held. The class requires a stated substitution only for first-band plants. For a second-band commodity, substitution came from other processors on the Department's encouragement, and the stop was short enough that it did not have to carry the load. The event does not test the requirement further.

6.7 §6.5 The Coordinated Cyber Scenario

Not exercised. One company was attacked, for money, and its process systems were not reached. What an adversary attacking several processors at once, at a moment of low stocks, could do is forward construction, offered to Project 9 (Section 10.2).

6.8 §7.3 The AI-Governance Design Basis

Not exercised. No component within the scope of *Layer Decomposition and Determinism* §4.3 bore on the event in the sources read.

6.9 Summary of Design Basis Performance

Of the eight readings, two Held, with a gap; three Held, the class's account among them; and three were Not exercised. Nothing was Missed. The class's §5.5 was written from this event and reads it. What the event adds is that the supply share that a cyber attack removes is set by the system the plants share, and that a short outage may not be credited to backups and core systems the next attack may reach.

7. Findings

Section references are to *Food Processing and Manufacturing* v0.1 unless another instrument is named.

7.1 F-1 — The Supply Share Is Counted Across the Shared System

An attack on servers serving one company's North American and Australian systems stopped all of its United States beef plants within a day, and national cattle slaughter fell 22 percent from a week earlier (Sections 3.2 and 3.3). The finding confirms §3.3's supply share and §5.5's consequence threshold, the stop of every plant that depends on the systems attacked, and is that the class should state the supply share for the cyber event of §5.5 as the share of every plant that depends on a common system whose loss would stop them, so that a company whose plants share business systems carries, for that event, the combined share of its plants; and that a company whose combined share places it in the second band states the substitution of §4.5 for that combined share.

Affected: §3.3, §4.5, §5.5. Disposition: R-1.

7.2 F-2 — Restoration Is Credited Only From What the Attack Cannot Reach

The company returned within days because its backups, encrypted and kept apart, were not infected, and because the attackers never reached its core systems (Section 3.4). The finding confirms §5.5's requirement of a manual mode and its refusal to release on untrusted records, and is that the class should state that a first- or second-band plant, or the company whose systems it depends on, credits restoration only from backups and systems shown to be beyond the reach

of an attack on its business network, with the time to restore from them stated and exercised; and that, under §6.9, a time to restart credited on the assumption that the core systems will not be reached is a relaxation requiring a recorded basis.

Affected: §3.3, §5.5, §6.9. Disposition: R-2.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 and R-2 dispose to *Food Processing and Manufacturing*.

No.	Title	Affected	Recommendation	Pri.
R-1	Count the supply share across the shared system	§3.3, §4.5, §5.5	For the cyber event, state the combined share of every plant sharing a system whose loss would stop them, and state substitution for that combined share.	1
R-2	Credit restoration only from what the attack cannot reach	§3.3, §5.5, §6.9	Credit only backups and systems shown beyond the business network's reach, with restoration time stated and exercised; treat a restart that assumes core systems survive as a relaxation needing a basis.	1

9. Open Items

OI-1: the register row. Project 0 is asked to enter the tier determination of Section 5.2. OI-2: the class has not taken in this instrument and conforms at its next revision; no erratum is owed. OI-3: a public body's figure for the company's share of beef capacity was not found; if one is, the class takes it in.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework, and makes no finding of attribution and none of fault.

10.2 Relationship to the Class's Use Case

The class's use case traces §6.3, deliberate contamination at the chokepoint timed to a denial of detection. This event bears on it through §6.5 and §4.4: it shows a company's plants losing their business systems in a day, which are the systems traceability and recall normally run on. The figures it supplies are a stop of every beef plant of one company within a day, a fall of about a fifth in national cattle slaughter, and a return in two to four days that rested on backups and core systems the attack did not reach. F-2 bears on what the use case may credit for the records from which a hold and recall would begin. An adversary attacking several processors at once, or their process systems, is the use case's own construction.

10.3 Relationship to Other Instruments

The Ransomware Attack on the Irish Health Service Executive (DBA-HPH-RE-1), *The Ransomware Attack on the Port of Nagoya* (DBA-TRAN-RE-2) and *The NotPetya Attack on Maersk's Terminals* (DBA-TRAN-RE-1) read attacks on business systems that stopped physical operations in other sectors.

References

JBS USA, *JBS USA Cyberattack Media Statement*, 31 May 2021; *JBS USA and Pilgrim's Announce Resolution of Cyberattack*, 3 June 2021; statement on the ransom payment, 9 June 2021. Federal Bureau of Investigation, statement on the JBS cyberattack, 2 June 2021. Bloomberg, report of 2 June 2021 on the stop of the company's United States beef plants, with the Department of Agriculture's slaughter estimates. The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 14 October 2026: issued in Project 11 under the Food and Agriculture Sector Framework v0.5, on the conferral of the framework's §9.1 of 14 October 2026, read against *Food Processing and Manufacturing* v0.1. Written new under the Reference Event Compression Standard v0.3, the series' first reference event instrument. Tier determination at Section 5.2; F-1 and F-2, R-1 and R-2. The class's account holds; no erratum is owed.